

MILESTONE-4 REPORT: EXOPLANET HABITABILITY PREDICTION

1. Project Overview

This project predicts exoplanet habitability using machine learning. It analyzes temperature, radius, mass, and distance to determine if a planet can support life. A web interface allows real-time prediction and visualization.

2. Theory of Code

The backend uses Python and Flask to handle requests. Data is processed using Pandas, where missing values are removed and key features are selected. A machine learning model (XGBoost/Random Forest) is trained and used to predict probability using `predict_proba()`. The frontend sends user input to backend and displays results like habitability %, closest planet, and Earth comparison.

3. System Architecture

Input → Data Processing → Machine Learning Model → Similarity Check → Output (Habitability %, Planet Match, Graphs)

4. Features

Real-time prediction, Earth comparison, closest planet match, interactive UI, and visual graphs.

5. Deployment

The system is deployed on Replit and accessible via web browser.

6. Application Link

<https://cosmic-habitation-explorer--ayushsingh05838.replit.app>

7. Application Interface



8. Conclusion

This project combines AI and astronomy to create an interactive tool for analyzing planetary habitability.
